

Program to Generate Message Digest Using SHA-1 and SHA-2

```
import hashlib

def get_digests(data: str):
    # Convert the input string to bytes
    byte_data = data.encode('utf-8')

    # SHA-1 Digest
    sha1 = hashlib.sha1(byte_data).hexdigest()

    # SHA-224 Digest (SHA-2 variant)
    sha224 = hashlib.sha224(byte_data).hexdigest()

    # SHA-256 Digest (SHA-2 variant)
    sha256 = hashlib.sha256(byte_data).hexdigest()

    # SHA-384 Digest (SHA-2 variant)
    sha384 = hashlib.sha384(byte_data).hexdigest()

    # SHA-512 Digest (SHA-2 variant)
    sha512 = hashlib.sha512(byte_data).hexdigest()

    return {
        'SHA-1': sha1,
        'SHA-224': sha224,
        'SHA-256': sha256,
        'SHA-384': sha384,
        'SHA-512': sha512
    }

if __name__ == "__main__":
    input_data = input("Enter the string to hash: ")
    digests = get_digests(input_data)
    for algo, digest in digests.items():
        print(f"{algo} digest: {digest}")
```